MATHS I · WEEK 1

Maths I week 1 formulas: Sets, Venn diagrams, relations and functions

Number systems & sets · Venn diagrams & counting · Relations · Functions. Save this PDF for revision.

NUMBER SYSTEMS & SETS · OFTEN ASKED

Number systems

Chain

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

IITM lectures take $\mathbb{N} = \{0, 1, 2, \ldots\}$. Some questions start at 1, so read the definition given.

Rational

$$p/q,\ p, q \in \mathbb{Z},\ q \neq 0$$

Decimal expansion terminates or recurs.

Irrational

$$\sqrt{2},\ \sqrt{3},\ \pi,\ e$$

Non-terminating, non-recurring. $\sqrt{n}$ is irrational unless n is a perfect square.

NUMBER SYSTEMS & SETS · OFTEN ASKED

Number systems

gcd × lcm

$$\gcd(a, b) \times \operatorname{lcm}(a, b) = a \times b$$

From prime factors

gcd = lowest powers of common primes · lcm = highest powers of all primes

NUMBER SYSTEMS & SETS · OFTEN ASKED

Sets (|A| = n)

Power set

$$|P(A)| = 2^n$$

Proper subsets

$$2^n - 1$$

Non-empty proper subsets: $2^n - 2$

Subsets of size k

$$C(n, k) = n!/(k!(n - k)!)$$

NUMBER SYSTEMS & SETS · OFTEN ASKED

Sets (|A| = n)

Cartesian product

$$|A \times B| = |A| \cdot |B|$$

$A \times B \neq B \times A$ unless $A = B$ or one of them is empty.

Difference

$$A \setminus B = A \cap B^c,\ |A \setminus B| = |A| - |A \cap B|$$

De Morgan

$$(A \cup B)^c = A^c \cap B^c \cdot (A \cap B)^c = A^c \cup B^c$$

NUMBER SYSTEMS & SETS · OFTEN ASKED

Key points

- ∅ is a subset of every set, and every set is a subset of itself.
- {∅} is not empty: it has one element.
- Repeats and order don't matter: {1, 2, 2} = {2, 1}.
- Set-builder: list it out before counting. {x ∈ ℤ | −2 < x ≤ 3} = {−1, 0, 1, 2, 3}.
- Intervals: [a, b] includes both ends, (a, b) excludes them.

NUMBER SYSTEMS & SETS · OFTEN ASKED

Common traps

⚠ $\in$ is for elements, $\subseteq$ is for sets: $1 \in \{1, 2\}$ but $\{1\} \subseteq \{1, 2\}$.

⚠ 0 is an integer and a rational number (0/1). It is neither positive nor negative.

⚠ 22/7 and 1.414 are rational; π and $\sqrt{2}$ are not.

VENN DIAGRAMS & COUNTING · IN EVERY PAPER

Venn diagrams & counting

Two sets

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Three sets

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

Only A

$$|A| - |A \cap B| - |A \cap C| + |A \cap B \cap C|$$

A and B, not C

$$|A \cap B| - |A \cap B \cap C|$$

Exactly two

$$(|A \cap B| + |B \cap C| + |A \cap C|) - 3|A \cap B \cap C|$$

VENN DIAGRAMS & COUNTING · IN EVERY PAPER

Venn diagrams & counting

Exactly one

$$|A| + |B| + |C| - 2(|A \cap B| + |B \cap C| + |A \cap C|) + 3|A \cap B \cap C|$$

Neither / none

$$\text{Total} - |A \cup B \cup C|$$

Largest overlap

$$|A \cap B|_{max} = \min(|A|, |B|)$$

Smallest overlap

$$|A \cap B|_{min} = \max(0, |A| + |B| - \text{Total})$$

VENN DIAGRAMS & COUNTING · IN EVERY PAPER

Common traps

⚠ Tip: Fill the diagram from the centre outwards: the triple overlap first, then the 'exactly two' regions, then the 'only' regions, then 'none' outside. Every sub-question is then just adding regions.

⚠ 'Both A and B' includes people who are also in C. 'Only A and B' does not.

⚠ 'At least one' is the union. 'Neither' is total − union.

⚠ If the total isn't given, never assume 'none' is 0.

RELATIONS · IN EVERY PAPER

Definitions (|A| = n)

Relation on A

$$R \subseteq A \times A$$

Number of relations

$$2^{(n^2)}$$

Reflexive

$$(a, a) \in R \text{ for every } a \in A$$

Number of reflexive relations: $2^{(n^2 - n)}$

Symmetric

$$(a, b) \in R \Rightarrow (b, a) \in R$$

Number of symmetric relations: $2^{(n(n + 1)/2)}$

RELATIONS · IN EVERY PAPER

Definitions (|A| = n)

Antisymmetric

$$(a, b) \in R \text{ and } (b, a) \in R \Rightarrow a = b$$

Transitive

$$(a, b) \in R \text{ and } (b, c) \in R \Rightarrow (a, c) \in R$$

Equivalence

reflexive + symmetric + transitive

Pairs not in R

$$|(A \times A) \setminus R| = n^2 - |R|$$

RELATIONS · IN EVERY PAPER

Standard relations

Relation	Refl.	Sym.	Anti.	Trans.
Identity {(a, a)}	✓	✓	✓	✓
Empty ∅ (A non-empty)	✗	✓	✓	✓
Universal A × A (|A| ≥ 2)	✓	✓	✗	✓
a ≤ b	✓	✗	✓	✓
a < b	✗	✗	✓	✓
a divides b (positive integers)	✓	✗	✓	✓
a − b is even (integers)	✓	✓	✗	✓
a ≠ b	✗	✓	✗	✗
Line ⊥ line	✗	✓	✗	✗

RELATIONS · IN EVERY PAPER

Common traps

⚠ Tip: Write R out as a list of pairs first (it's usually under 15 pairs), then test each property against that list. Reasoning from the rule alone is where marks are lost.

⚠ Transitivity trap: if (1, 2) and (2, 1) are in R, then (1, 1) and (2, 2) must be in R too.

⚠ Reflexive needs (a, a) for every element of A, including elements that never appear in R.

RELATIONS · IN EVERY PAPER

Common traps

⚠ Symmetric and antisymmetric aren't opposites. The identity relation is both; {(1, 2), (2, 1), (1, 3)} is neither.

⚠ No chain (a, b), (b, c) at all? Then R is transitive (vacuously). ∅ is symmetric and transitive.

FUNCTIONS · OFTEN ASKED

|A| = m, |B| = n

Function f: A → B

every a ∈ A has exactly one image in B

Range

{f(a) : a ∈ A} ⊆ codomain B

One-one (injective)

$f(a_1) = f(a_2) \Rightarrow a_1 = a_2$

Onto (surjective)

range = codomain

Bijective

one-one + onto ⇒ f^(–1) exists

FUNCTIONS · OFTEN ASKED

|A| = m, |B| = n

Functions A → B

$$n^m$$

One-one functions (m ≤ n)

$$n!/((n - m)!)$$

Bijections (m = n)

$$n!$$

Composition

$$(g \circ f)(x) = g(f(x))$$

Usually $g \circ f \neq f \circ g$.

FUNCTIONS · OFTEN ASKED

Tests and domains

Check	Rule
Vertical line test	Every vertical line meets the graph at most once ⇒ it is a function
Horizontal line test	Every horizontal line meets the graph at most once ⇒ one-one
$\sqrt{g(x)}$	Domain: $g(x) \geq 0$
$1/g(x)$	Domain: $g(x) \neq 0$
$\log g(x)$	Domain: $g(x) > 0$

FUNCTIONS · OFTEN ASKED

Standard functions, $\mathbb{R} \to \mathbb{R}$

f(x)	One-one?	Onto?
$ax + b$ $(a \neq 0)$	✓	✓
x^2	✗	✗ (range $[0, \infty)$)
x^3	✓	✓
$\lvert x \rvert$	✗	✗
e^x	✓	✗ (range $(0, \infty)$)
x^2 on $[0, \infty) \to [0, \infty)$	✓	✓

FUNCTIONS · OFTEN ASKED

Common traps

⚠ Changing the domain or codomain changes the answer: x^2 is not one-one on $\mathbb{R}$, but is on $[0, \infty)$.

⚠ If some element of A has no image, or two images, it is not a function at all.

⚠ Onto depends on the codomain the question states, not on the formula.

Notes for every week of Maths I, free.

Full notes, the formula sheet and common traps, with worked PYQs.

btechi.in/qualifier/notes/maths-1 →

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